

R Programming Notes

Data Objects, Vectors, Lists, Arrays, Matrices, Functions and Data
Frames

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1 Data Objects

1.1 Syntax

```
1 # Numeric Object
2 x <- 25
3
4 # Character Object
5 name <- "R Programming"
6
7 # Logical Object
8 flag <- TRUE
9
10 # Complex Object
11 z <- 3 + 2i
```

1.2 Operations

```
1 print(x)
2 class(x)
3 typeof(x)
4 length(x)
5 rm(x)
```

2 Vectors

2.1 Creating Vectors

```
1 v1 <- c(10,20,30,40,50)
2
3 v2 <- 1:10
4
5 v3 <- seq(5,25,5)
6
7 v4 <- rep(2,5)
8
9 v5 <- rep(c(1,2,3),3)
```

2.2 Accessing Elements

```
1 v1[1]
2
3 v1[3]
4
5 v1[c(2,4)]
6
```

```
7 v1[2:5]
8
9 v1[-3]
10
11 v1[length(v1)]
```

2.3 Modifying Elements

```
1 v1[2] <- 100
2
3 v1[5] <- 500
4
5 v1
```

2.4 Vector Operations

```
1 v1 + 5
2
3 v1 * 2
4
5 v1 + c(1,2,3,4,5)
6
7 v1 - c(1,1,1,1,1)
8
9 v1 / 10
10
11 v1 ^ 2
```

2.5 Useful Vector Functions

```
1 length(v1)
2
3 sum(v1)
4
5 mean(v1)
6
7 max(v1)
8
9 min(v1)
10
11 sort(v1)
12
13 rev(v1)
```

2.6 Logical Operations

```

1 v1 > 20
2
3 v1[v1 > 20]
4
5 v1[v1 %% 2 == 0]

```

2.7 Syntax

```

1 vector_name <- c(value1, value2, value3, ...)

```

2.8 Example

```

1 marks <- c(75, 82, 91, 68, 87)
2
3 print(marks)

```

2.9 Operations

```

1 marks[1]           # First element
2 marks[3]           # Third element
3 marks[2:4]         # Elements 2 to 4
4 marks[-2]          # Remove second element
5 length(marks)
6 sum(marks)
7 mean(marks)
8 max(marks)
9 min(marks)
10 sort(marks)

```

3 Lists

3.1 Creating Lists

```

1 student <- list(
2   Name="Arun",
3   Age=20,
4   CGPA=8.7,
5   Passed=TRUE
6 )

```

3.2 Accessing List Elements

```
1 student$Name
2
3 student$Age
4
5 student[[3]]
6
7 student[1]
```

3.3 Modifying Lists

```
1 student$Age <- 21
2
3 student$Department <- "CSE"
4
5 student
```

3.4 Syntax

```
1 list_name <- list(
2 item1 = value1,
3 item2 = value2,
4 ...
5 )
```

3.5 Example

```
1 student <- list(
2 Name = "Arun",
3 Age = 20,
4 CGPA = 8.9,
5 Passed = TRUE
6 )
7
8 print(student)
```

3.6 Operations

```
1 student$Name
2 student$Age
3 student[[3]]
4 student$Passed
```

4 Matrices

4.1 Creating Matrices

```
1 A <- matrix(1:9,nrow=3)
2
3 B <- matrix(1:12,nrow=3,
4 byrow=TRUE)
```

4.2 Accessing Matrix Elements

```
1 A[1,2]
2
3 A[3,1]
4
5 A[2,]
6
7 A[,3]
8
9 A[1:2,2:3]
```

4.3 Matrix Operations

```
1 A+B
2
3 A-B
4
5 A*2
6
7 A%*%A
8
9 t(A)
10
11 det(A)
12
13 solve(A)
```

4.4 Syntax

```
1 matrix_name <- matrix(
2 data,
3 nrow = rows,
4 ncol = columns,
5 byrow = TRUE/FALSE
6 )
```

4.5 Example

```
1 A <- matrix(1:9,  
2 nrow = 3,  
3 byrow = TRUE)  
4  
5 print(A)
```

4.6 Operations

```
1 A[1,2]  
2 A[2,]  
3 A[,3]  
4 t(A)  
5 A %*% A
```

5 Arrays

5.1 Creating Arrays

```
1 arr <- array(1:12,  
2 dim=c(3,2,2))
```

5.2 Accessing Array Elements

```
1 arr[1,2,1]  
2  
3 arr[, ,1]  
4  
5 arr[,1,]  
6  
7 arr[2,,]
```

6 Multi-dimensional Arrays

6.1 Creating Multi-dimensional Arrays

```
1 multi <- array(  
2 1:48,  
3 dim=c(2,3,4,2)  
4 )
```


6.2 Accessing Elements

```
1 multi[1,2,3,1]
2
3 multi[,,,2]
4
5 multi[1,,,]
```

6.3 Syntax

```
1 matrix_name <- matrix(
2 data,
3 nrow = rows,
4 ncol = columns,
5 byrow = TRUE/FALSE
6 )
```

6.4 Example

```
1 A <- matrix(1:9,
2 nrow = 3,
3 byrow = TRUE)
4
5 print(A)
```

6.5 Operations

```
1 A[1,2]
2 A[2,]
3 A[,3]
4 t(A)
5 A %*% A
```

7 Functions

7.1 Creating Functions

```
1 square <- function(x)
2 {
3   return(x*x)
4 }
```

7.2 Calling Functions

```
1 square(5)
```

7.3 Function with Multiple Arguments

```
1 add <- function(a,b)
2 {
3   return(a+b)
4 }
5
6 add(10,20)
```

7.4 Default Arguments

```
1 power <- function(x,n=2)
2 {
3   return(x^n)
4 }
5
6 power(5)
7
8 power(5,3)
```

8 Data Frames

8.1 Creating Data Frames

```
1 students <- data.frame(
2   ID=c(101,102,103),
3   Name=c("Arun","Bala","Divya"),
4   Marks=c(80,91,75)
5 )
```

8.2 Viewing Data

```
1 students
2
3 head(students)
4
5 tail(students)
6
7 str(students)
8
9 summary(students)
```

8.3 Accessing Columns

```
1 students$Marks
2
3 students[,2]
4
5 students[1,]
6
7 students[2,3]
```

8.4 Selecting Rows

```
1 students[students$Marks>80,]
2
3 students[students$Marks<80,]
```

8.5 Adding Columns

```
1 students$Grade <- c("A","A","B")
```

8.6 Adding Rows

```
1 newrow <- data.frame(
2 ID=104,
3 Name="Kiran",
4 Marks=88,
5 Grade="A"
6 )
7
8 students <- rbind(
9 students,
10 newrow
11 )
```

8.7 Deleting Columns

```
1 students$Grade <- NULL
```

8.8 Useful Data Frame Functions

```
1 nrow(students)
2
3 ncol(students)
4
5 names(students)
6
```

```
7 dim(students)
8
9 colnames(students)
```

8.9 Syntax

```
1 data.frame(
2   column1 = ...,
3   column2 = ...
4 )
```

8.10 Example

```
1 students <- data.frame(
2   ID = c(101,102,103),
3   Name = c("Arun","Bala","Divya"),
4   Marks = c(85,91,76)
5 )
6
7 print(students)
```

8.11 Operations

```
1 students$Marks
2 students[,2]
3 students[students$Marks > 80,]
4 students$Grade <- c("A","A","B")
```

9 Common Vectorized Operations

```
1 x <- c(10,20,30,40)
2
3 y <- c(1,2,3,4)
4
5 x+y
6
7 x-y
8
9 x*y
10
11 x/y
12
13 x>20
14
15 x[x>20]
```

```
16
17 sum(x)
18
19 mean(x)
20
21 max(x)
22
23 min(x)
24
25 sort(x)
```

10 Useful Built-in Functions

```
1 length()
2
3 class()
4
5 typeof()
6
7 sum()
8
9 mean()
10
11 median()
12
13 min()
14
15 max()
16
17 sort()
18
19 seq()
20
21 rep()
22
23 c()
24
25 matrix()
26
27 array()
28
29 data.frame()
30
31 list()
32
33 factor()
34
35 dim()
36
```

```
37 nrow()  
38  
39 ncol()  
40  
41 head()  
42  
43 tail()  
44  
45 summary()  
46  
47 str()
```